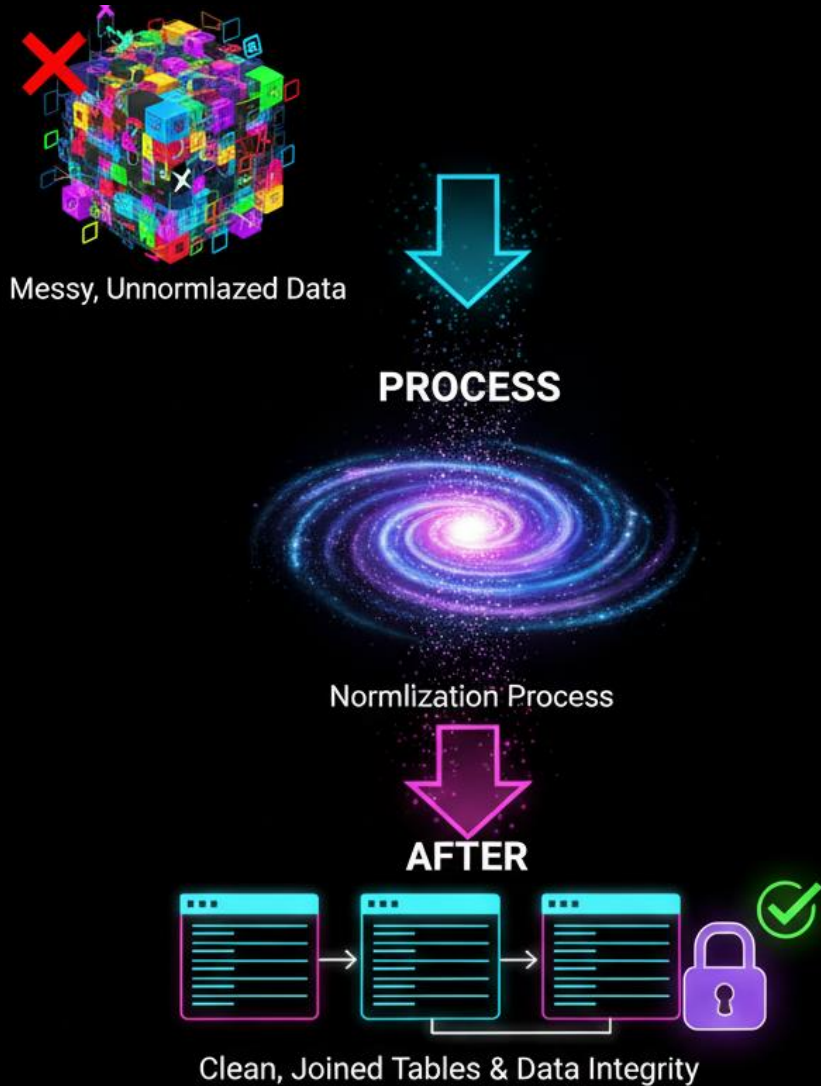


Foundations & Basics

Normalization and Data Integrity



By: Zherou A. Majied

Database Normalization: Organizing Data

- **Normalization**

Is the process of organizing data in database to reduce redundancy and avoid update anomalies.

- List of Normal Forms

1. **1NF**: Remove repeating groups & multi-valued columns
2. **2NF**: Remove partial dependencies
3. **3NF**: Remove transitive dependencies
4. ~~**BCNF**: Every determinant must be a candidate key~~
5. ~~**4NF**: Remove multi-valued dependencies~~
6. ~~**5NF**: Break down tables to remove join dependencies, very advanced~~
7. ~~**6NF**: Break tables down to the point where temporal (time-based) data is perfectly handled~~

ONF: Unnormalized Form

ONF									
Name	Addr	Email	Phone	Orders	Products	Quantities	Prices	Brand	Country
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Pen, Notebook	2, 1	1.5, 3.00	Bic, Moleskine	USA, Italy
Areen Osman	10 Salem St.	areen@xyz.co	256-5845	102	Whiteboard	1	35.00	Quartet	USA
Areen Osman	10 Salem St.	areen@xyz.co	256-5845	103	Pencil, Notebook	16, 2	0.5, 2.75	Dixon, Moleskine	USA, Italy

Problems

1. **Repeating groups**
 - Multiple Products, Quantities, Prices, etc. in one cell
2. No clear unique key for individual items
3. Hard to query, update, or maintain
4. **Data redundancy:**
 - Customer details repeated for every order

1NF: Remove Repeating Groups & Multi-Valued Columns

- Each column must have atomic values
- No repeating groups, one fact per row

Benefit

Now we can uniquely identify each row with (OrderID, Product).

Issues:

- Redundant customer info across multiple rows
- Modification anomalies
- Partial dependency
- Transitive dependency

ONF									
Name	Addr	Email	Phone	Orders	Products	Quantities	Prices	Brand	Country
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1NF									
Name	Addr	Email	Phone	Orders	Products	Quant.	Prices	Brand	Country
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Pen	2	1.5	Bic	USA
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Notebook	1	3.00	Moleskine	Italy
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Areen Osman	10 Salem St.	areen@xyz.co	256-5845	103	Notebook	2	2.75	Moleskine	Italy

Data Anomalies



Data anomalies are problems that occur when data is stored redundantly or improperly, making **updates**, **inserts**, or **deletes** inconsistent

Name	Addr	Email	Phone	Orders	Products	Quant.	Prices	Brand	Country
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Pen	2	1.5	Bic	USA
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Notebook	1	3.00	Moleskine	Italy
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Update Anomaly	If Areen phone number changes, we must update it in every row
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Insert Anomaly	A brand cannot be added unless there is an actual order for it
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Delete Anomaly	If order 102 is delete, we looks Whiteboard product
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Functional Dependency (FD)

A **functional dependency (FD)** is a general rule that defines how one column or set of columns determines another column in the same table.

- **Partial Dependency**

- A dependency where only part of a composite primary key determines another column

- **Transitive Dependency**

- A dependency where a non-key column depends on another non-key column, rather than directly on the primary key

Name	Addr	Email	Phone	Orders	Products	Quantities	Prices	Brand	Country
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Pen	2	1.5	Bic	USA
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Notebook	1	3.00	Moleskine	Italy
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2NF: Remove Partial Dependencies

- Must already be in 1NF
- Remove partial dependencies:
 - No non-key attribute should depend on part of a composite key

1NF									
Name	Addr	Email	Phone	Orders	Products	Q.	Prices	Brand	Country
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Pen	2	1.5	Bic	USA
Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234	101	Notebook	1	3.00	Moleskine	Italy
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Areen Osman	10 Salem St.	areen@xyz.co	256-5845	103	Notebook	2	2.75	Moleskine	Italy

Customer				
Cust_ID	Name	Addr	Email	Phone
1	Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234
2	Areen Osman	10 Salem St.	areen@xyz.co	256-5845

Orders	
Order_ID	Cust_ID
101	1
102	2
103	2

Order_Details			
Order_ID	Product_ID	Quantities	Prices
101	1	2	1.5
101	2	1	3.00
102	3	1	35.00
103	4	16	0.5
103	2	2	2.75

Products			
Product_ID	P_Name	Brand	Country
1	Pen	Bic	USA
2	Notebook	Moleskine	Italy
3	Whiteboard	Quartet	USA
4	Pencil	Dixon	USA

Issues:

1. ~~Redundant customer info across multiple rows~~
2. ~~Modification anomalies~~
3. ~~Partial dependency~~
4. Transitive dependency

3NF: Remove Transitive Dependencies

- Must already be in 2NF
- Remove transitive dependencies:
 - Non-key attributes must depend only on the primary key, not on other non-key attributes.

Products

Product_ID	P_Name	Brand	Country
1	Pen	Bic	USA
2	Notebook	Moleskine	Italy
3	Whiteboard	Quartet	USA
4	Pencil	Dixon	USA

Products

Product_ID	Products	Brand_ID
1	Pen	1
2	Notebook	2
3	Whiteboard	3
4	Pencil	4

Brands

Brand_ID	B_Name	B_Country
1	Bic	USA
2	Moleskine	Italy
3	Quartet	USA
4	Dixon	USA

Unnormalized vs. Normalized Data

Name	Addr	Email	Phone	Orders	Products	Quantities	Prices	Brand	Country
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Customer

Cust_ID	Name	Addr	Email	Phone
1	Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234
2	Areen Osman	10 Salem St.	areen@xyz.co	256-5845

Orders

Order_ID	Cust_ID
101	1
102	2
103	2

Order_Details

Order_ID	Product_ID	Quantities	Prices
101	1	2	1.5
101	2	1	3.00
102	3	1	35.00
103	4	16	0.5
103	2	2	2.75

Brands

Brand_ID	B_Name	B_Country
1	Bic	USA
2	Moleskine	Italy
3	Quartet	USA
4	Dixon	USA

Products

Product_ID	Products	Brand_ID
1	Pen	1
2	Notebook	2
3	Whiteboard	3
4	Pencil	4

Denormalization: When We Combine Tables for Performance

- **Denormalization:**

is the process of combining normalized tables back into fewer tables to improve query performance, often at the cost of data redundancy

Orders

Order_ID	Cust_ID
101	1
102	2
103	2

Customer

Cust_ID	Name	Addr	Email	Phone
1	Rawaz Zrar	985 Oak St.	rawaz.z@exe.org	555.1234
2	Areen Osman	10 Salem St.	areen@xyz.co	256-5845

Order_Details

Order_ID	Product_ID	Quantities	Prices
101	1	2	1.5
101	2	1	3.00
102	3	1	35.00
103	4	16	0.5
103	2	2	2.75

Products

Product_ID	Products	Brand_ID
1	Pen	1
2	Notebook	2
3	Whiteboard	3
4	Pencil	4

Brands

Brand_ID	B_Name	B_Country
1	Bic	USA
2	Moleskine	Italy
3	Quartet	USA
4	Dixon	USA

Denormalized Table

Name	Orders	Products	Quantities	Prices
Rawaz Zrar	101	Pen	2	1.5
Rawaz Zrar	101	Notebook	1	3.00
Areen Osman	102	Whiteboard	1	35.00
Areen Osman	103	Pencil	16	0.5
Areen Osman	103	Notebook	2	2.75

